= 1 , 2 , 3 , 4: four independent equations.

4. (I) The satellite clock slows by $[1-(v/c)^2]^{1/2}$. The ground reading T minus satellite reading t is

$$T-t=T-\sqrt{1-(\frac{v}{c})^2}\,T=[\,1-\sqrt{1-(\frac{v}{c})^2}\,]\,T\ , \qquad (11)$$

Here v is satellite speed relative to the ground, determined by

$$\frac{v^2}{r}=\frac{GM}{r^2}\ , \qquad (12)$$

G is the gravitational constant, M Earth's mass, r orbit radius. From (11),

$$v=\sqrt{\frac{GM}{r}}=\sqrt{\frac{g}{r}}R=\sqrt{\frac{g}{R+h}}R\ ,$$

Here R is Earth's radius, h altitude and $g=GM/R^2$ ground gravity. Substitution gives $v=$ 3489 , hence $(v/c)^2\approx$ 1.68 $^{-10}$, which is very small. Thus

$$[\,1-(\frac{v}{c})^2\,]^{1/2}\approx 1-\frac{1}{2}(\frac{v}{c})^2\ .$$

Finally the clock difference over 24 is

$$T-t\approx\frac{1}{2}(\frac{v}{c})^2T=\frac{1}{2}\,\frac{gR^2}{c^2(R+h)}T=7.3\,\mu\text{s}\ . \qquad (13)$$

(II) The satellite reading t minus the inertial-clock reading T_0 at infinity is

$$t-T_0=\sqrt{1-2\frac{\phi}{c^2}}T_0-T_0=(\sqrt{1-2\frac{\phi}{c^2}}-1)T_0\ . \qquad (14)$$

The gravitational potential magnitude at the satellite clock is

$$\phi=\frac{GM}{R+h}=\frac{R^2}{R+h}g\ . \qquad (15)$$

Thus $\dfrac{\phi}{c^2}=\dfrac{gR^2}{c^2(R+h)}$;

Numerically, $\phi/c^2=$ 1.68 $^{-10}$ is very small, so equation (14) becomes

$$t-T_0\approx-\frac{\phi}{c^2}T_0\ . \qquad (16)$$

Similarly, ground reading T minus the inertial clock at infinity is

$$T-T_0=\sqrt{1-2\frac{\phi_E}{c^2}}T_0-T_0=(\sqrt{1-2\frac{\phi_E}{c^2}}-1)T_0\ . \qquad (17)$$

The gravitational potential magnitude at the ground clock is

$$\phi_E=\frac{GM}{R}=gR\ . \qquad (18)$$